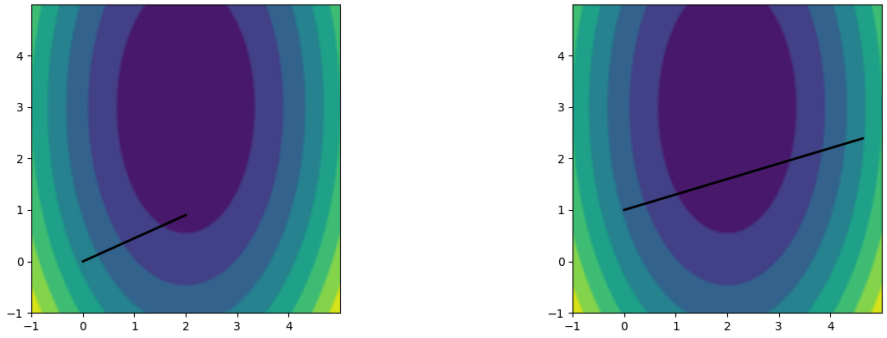
Discussion: We consider the gradient descent method to solve optimization problems. In the following figure, the first step of a gradient descent method with update rule

$$x^{k+1} = x^k - \alpha \nabla f(x^k)$$

for an arbitrary $\alpha > 0$ are visualized for different initial iterates x^0 (start and endpoint of the line correspond to x^0 and x^1). Argue which of the points was the initial iterate.



Exercise 4.1

We consider the function

$$f : \mathbb{R}^2 \to \mathbb{R}, \quad f(x) := (x_1 - 3)^2 + k(x_0 - 2)^2$$

for $k = 10$.

1. Show f is L-smooth and determine L.
2. Implement the gradient descent method with update rule

$$x^{k+1} = x^k - \alpha_k \nabla f(x^k), \tag{1}$$

for step length rule $\alpha_k = \frac{1}{L}$ for all $k \in \mathbb{N}$. where L denotes the Lipschitz constant of the gradient of f. Terminate the loop as soon as $k \geq$ `maxiter` or $\|\nabla f(x^k)\|_2 \leq$ `epsilon`, with `maxiter` $= 1000$ and `epsilon` $= 10^{-5}$. Experiment with other step lengths and conclude.

Exercise 4.2

Let $f : \mathbb{R}^n \to \mathbb{R}$ be twice continuously differentiable and m-strongly convex. Let $x^0 \in \mathbb{R}^n$, an initial point and $S = \{x \in \text{dom} f, f(x) \leq f(x^0)\}$. Assume that there exist constants $0 < m \leq M$ such that

$$mI \preceq \nabla^2 f(x) \preceq MI, \qquad \forall x \in S.$$

Let $x^\star$ denote the unique minimizer of f.
Consider the gradient descent method with exact line search:

$$x^{k+1} = x^k - \alpha_k \nabla f(x^k), \qquad \alpha_k \in \arg\min_{\alpha \geq 0} f\big(x^k - \alpha \nabla f(x^k)\big).$$

1. Show that
$$f(x^k) - f(x^\star) \leq \Big(1 - \frac{m}{M}\Big)^k \big(f(x^0) - f(x^\star)\big).$$
.

2. Let $\varepsilon > 0$. Determine the minimum number of iterations k needed to guarantee
$$f(x^k) - f(x^\star) \leq \varepsilon.$$

3. Discuss whether this method is computationally efficient in practice. In particular, comment on the cost of computing α_k at each iteration. Propose a practical alternative for choosing the step size.

Exercise 4.3

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^4$ be the dataset defined by

$$x_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad y_1 = 0, \qquad x_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad y_2 = 1, \qquad x_3 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad y_3 = 1, \qquad x_4 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad y_4 = 0.$$

Consider the function $f : \Theta \times \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(\theta, x) = \sigma\big(W^{(2)}\, \sigma\big(W^{(1)} x + b^{(1)}\big) + b^{(2)}\big),$$

where $\theta = \big(W^{(1)}, b^{(1)}, W^{(2)}, b^{(2)}\big)$, with, $W^{(1)} \in \mathbb{R}^{2\times 2}$, $b^{(1)} \in \mathbb{R}^2$, $W^{(2)} \in \mathbb{R}^{1\times 2}$, $b^{(2)} \in \mathbb{R}$ and $\sigma : \mathbb{R} \to \mathbb{R}$ is the sigmoid function

$$\sigma(t) = \frac{1}{1 + e^{-t}},$$

applied componentwise. For $(x, y) \in \mathbb{R}^2 \times \mathbb{R}$, define the loss

$$\ell(\theta, x, y) = \frac{1}{2}\big(f(\theta, x) - y\big)^2.$$

The empirical objective function, assuming that the dataset $\mathcal{D}$ is fixed, is

$$L(\theta) = \frac{1}{4} \sum_{i=1}^4 \ell(\theta, x_i, y_i).$$

1. For a fixed pair (x, y), give the gradients
$$\nabla_{W^{(2)}} \ell(\theta, x, y), \qquad \nabla_{b^{(2)}} \ell(\theta, x, y), \qquad \nabla_{W^{(1)}} \ell(\theta, x, y), \qquad \nabla_{b^{(1)}} \ell(\theta, x, y).$$

2. Implement a program that has the following functions

(a) `Forward`: Returns $f(\theta, x)$,

(b) `Empirical Loss`

(c) `Backward`: Returns $\nabla L(\theta)$. (make use of .backward() if using pytorch autograd)

(d) `Train`: runs gradient descent algorithm to minimize $L(\theta)$.

$$\theta^{k+1} = \theta^k - \eta \nabla L(\theta^k), \qquad k \in \mathbb{N},$$

where $\eta > 0$ is the step size, for $T \in \mathbb{N}$ iterations.

3. Test different step size choices η and plot the sequence $\left(L(\theta^k)\right)_{k=0}^{T}$.